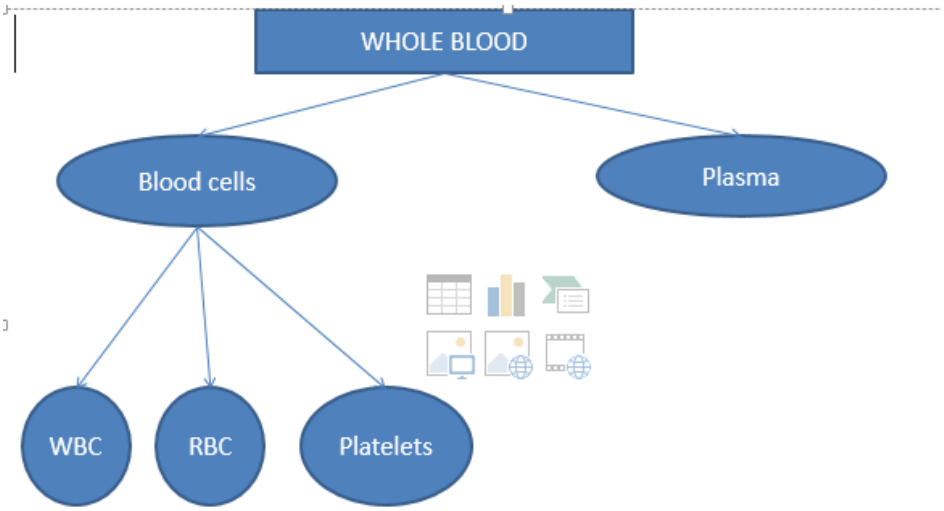
BLOOD

Blood is a connective tissue in fluid form.

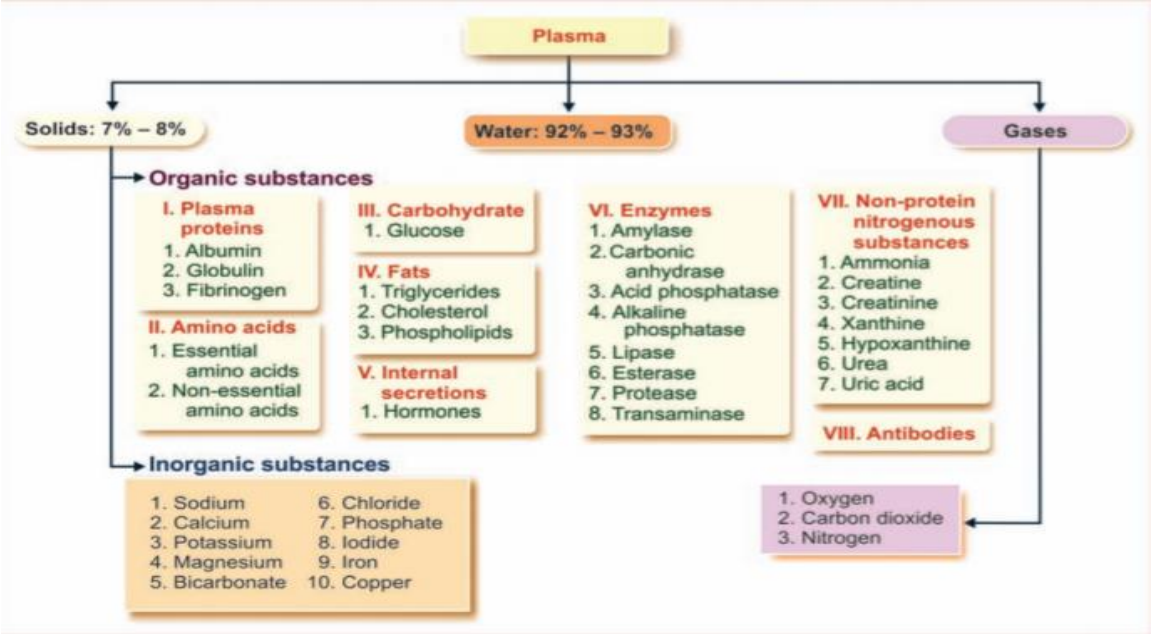
It is considered as:

- Fluid of life
- Fluid of growth
- Fluid of health

COMPOSITION OF BLOOD



COMPOSITION OF PLASMA



Ms. SNEHA BABY
LECTURER
DEPARTMENT OF PHYSIOLOGY

1. NUTRITIVE FUNCTION:

Nutritive substances like glucose, amino acids, lipids and vitamins derived from digested food are absorbed from gastrointestinal tract and carried by blood to different parts of the body for growth and production of energy.

2. RESPIRATORY FUNCTION:

Transport of respiratory gases is done by the blood. It carries oxygen from alveoli of lungs to different tissues and carbon dioxide from tissues to alveoli.

3. EXCRETORY FUNCTION

Waste products formed in the tissues during various metabolic activities are removed by blood and carried to the excretory organs like kidney, skin, liver, etc.

4. TRANSPORT OF HORMONES AND ENZYMES

Hormones which are secreted by ductless (endocrine) glands are released directly into the blood. The blood transports these hormones to their target organs/tissues.

Blood also transports enzymes.

5. STORAGE FUNCTION

- Blood serves as a readymade source for the substances like water, protiens, glucose, Na+, K+.
- These substances are taken from blood during the conditions like starvation, fluid loss, electrolyte loss, etc.

6. DEFENSIVE FUNCTION

- The white blood cells are responsible for this function.
- Neutrophils and monocytes engulf the bacteria by phagocytosis.
- Lymphocytes are involved in development of immunity.
- Eosinophils are responsible for detoxification, disintegration and removal of foreign proteins.

7. REGULATION OF WATER BALANCE

8. REGULATION OF ACID BASE BALANCE

9. REGULATIONOF BODY TEMPERATURE

PROPERTIES OF BLOOD

COLOUR:

Arterial blood- scarlet red

Venous blood- purple red

Arterial blood v/s venous blood

VOLUME:

Average volume of normal adult is 5L

pH:

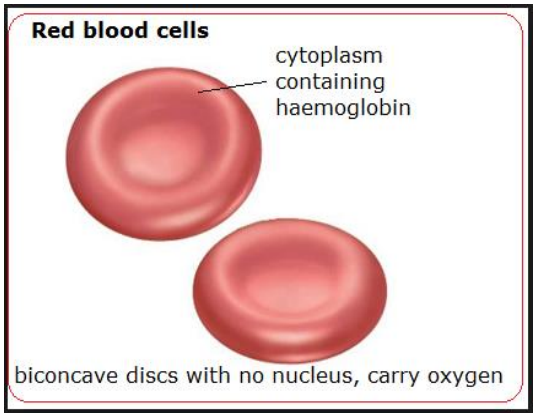
Blood is slightly alkaline and its pH is 7.4 in normal conditions.

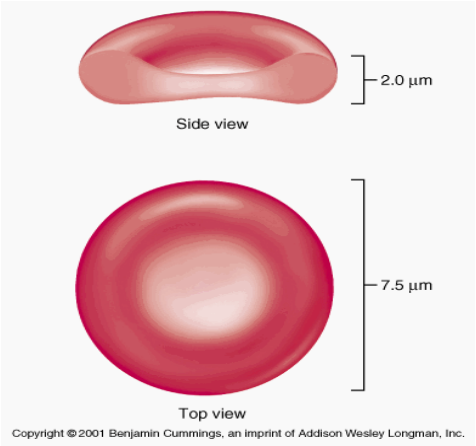
VISCOSITY:

Blood is 5 times more viscous than water

RED BLOOD CELL (RBC)

- Red blood cells (RBCs) are the non-nucleated formed elements in the blood.
- It is also known as erythrocytes (erythros = red).
- Red color of the red blood cell is due to the presence of the coloring pigment called hemoglobin.
- RBCs are larger in number compared to the other two blood cells, namely white blood cells and platelets.





Normal size:
- Diameter- 7.2µm
- Thickness- Centre: 1µm

  Periphery: 2µm
- Surface area- 120-140µm
- Volume- 80µm (78-86µm)

Normal shape:
- Circular
- Biconcave

Ms. SNEHA BABY
LECTURER
DEPARTMENT OF PHYSIOLOGY

Advantages of having biconcave shape:

- While passing through minute capillaries, RBCs squeeze through the capillaries very easily without getting damaged.
- It increases the surface area of RBC for the diffusion of respiratory gases.

Advantages of having no nucleus:

- It can accommodate more haemoglobin.

Normal count:

Clinically a count of 5million/cu mm is considered as 100%.

In males: 5million/cu mm

In females: 4.5 million/cu mm

Variation in size:

Variation in size is called ***Anicocytosis.***

2 types: Microcytocis

Macrocytosis

Variations in shape:

- Variation in shape is called ***Poikilocytosis.***
- Abnormal shapes are:
- Spherocyte
- Elliptocytes
- Sickle cell

Variation in count:

I. Physiolgical variation
II. Pathological variation

I. Physiological increase in the RBC count

1. Age :

At birth the RBC count is 6−7 million/µL.

(After about 10 days of birth the count decreases due to the destruction of cells. This is the cause of physiological jaundice of newborn.)

In infants

RBC count is slightly more than the adults.

2. Sex:

Adult females (average 4.5 million/µL)

Adult males (average 5.5 million/µL).

3. **High altitude**

Individuals residing in high altitude areas (above 10,000 feet from the sea level) have high RBC

count (7million/µL) because of the hypoxic stimulation of erythropoiesis.

4. Excessive exercise

Mild hypoxia and spleen contraction causes temporary increase in the RBC count.

5. Emotional conditions

Like anxiety are associated with the temporary increase in the RBC count due to sympathetic stimulation.

6. After meals

The RBC count is raised slightly.

II. Pathological increase in the RBC count

Polycythemia (above 7 million/µL)

Physiological decrease in RBC count is seen in the following conditions:

- At high barometric pressure,
- After sleep and
- In pregnancy.

Pathological decrease in RBC count

- Anaemia

(In anaemia, there may occur marked reduction in the RBC count or the Hb level or both)

ERYTHROPOIESIS

Erythropoiesis is the process of the origin, development and maturation of erythrocytes.

Stage	Duration	Site
Mesoblastic stage	1st two months of intra uterine life	Yolk sac
Hepatic stage	From 3rd month of intrauterine life	Liver and spleen
Myeloid stage	Last 3 months f intrauterine life	Red bone marrow and liver

DEVELOPEMENT

Pluripotent Hematopoietic stem cell

↓

Pronormoblast (proerythroblast)

↓

Early normoblast

↓

Intermediate normoblast

↓

Late normoblast

↓

Reticulocyte

↓ Vitamin 12

Mature RBC

Duration

Differentiation phase: from pronormoblast to reticulocyte phase - 5 days

Maturation phase: from reticulocyte to RBC - 2 days

Factor needed of Erythropoiesis

Development and maturation of erythrocytes require variety of factors, which are classified into three categories:

1. General factors
2. Maturation factors
3. Factors necessary for hemoglobin formation.

1. GENERAL FACTORS

General factors necessary for erythropoiesis are:

- Erythropoietin
- Thyroxine
- Hemopoietic growth factors
- Vitamins.

Erythropoietin

- Most important general factor for erythropoiesis is the hormone called erythropoietin.
- It is also called hemopoietin or erythrocyte stimulating factor.

- The erythropoietin is secreted by peritubular capillaries of kidney.
- Small quantity is also secreted from liver and brain.
- After the formation it is released into the blood.

Stimulating factor: Hypoxia

Thyroxine

Thyroxine accelerates the process of erythropoiesis at many levels.

Hemopoietic Growth Factors

Hemopoietic growth factors or growth inducers are the interleukins and stem cell.

Generally these factors induce the proliferation of PHSCs.

Vitamins

- Vitamin B: Its deficiency causes anemia and pellagra
- Vitamin C: Its deficiency causes anemia and scurvy
- Vitamin D: Its deficiency causes anemia and rickets
- Vitamin E: Its deficiency leads to anemia and malnutrition

2. **MATURATION FACTORS**

- **Vitamin B12**
- **Intrinsic factor**
- **Folic acid**

<u>Vitamin B12 (Cyanocobalamine)</u>

- Vitamin B12 is called extrinsic factor since it is obtained mostly from diet.
- Its absorption from intestine requires the presence of intrinsic factor of Castle.
- Vitamin B12 is stored mostly in liver and in small quantity in muscle.
- When necessary, it is transported to the bone marrow to promote maturation of RBCs.
- It is also produced in the large intestine by the intestinal flora.

Action:

- Vitamin B12 is essential for synthesis of DNA in RBCs.
- Its deficiency leads to failure in maturation of the cell and reduction in the cell division.
- Also, the cells are larger with fragile and weak cell membrane resulting in macrocytic anemia.
- Deficiency of vitamin B12 causes pernicious anemia.

<u>Intrinsic factor of castle</u>

- It is produced in gastric mucosa by the parietal cells of the gastric glands.
- It is essential for the absorption of vitamin B12 from intestine.

- In the absence of intrinsic factor, vitamin B12 is not absorbed from intestine.
- This leads to pernicious anemia.
- Deficiency of intrinsic factor occurs in:
  - ✓ Severe gastritis
  - ✓ Ulcer
  - ✓ Gastrectomy.

Folic Acid

- It is required for the synthesis of DNA.
- In the absence of folic acid, the synthesis of DNA decreases causing failure of maturation.
- This leads to anemia in which the cells are larger and appear in megaloblastic (proerythroblastic) stage.
- And, anemia due to folic acid deficiency is called megaloblastic anemia.

3. FACTORS NECESSARY FOR HEMOGLOBIN FORMATION

- Iron: Necessary for the formation of heme part of the hemoglobin.
- Copper: Necessary for the absorption of iron from the gastrointestinal tract.
- Cobalt and nickel: These metals are essential for the utilization of iron during hemoglobin formation.
- Vitamins: Vitamin C, riboflavin, nicotinic acid and pyridoxine are also essential for the formation of hemoglobin.

HAEMOGLOBIN

Definition:

- It is an iron containing colouring matter which is present on the cytoplasm of RBC's.
- It is the erythrocyte precursers synthesize the Hb while the mature erythrocyte lose the property of synthesizing Hb.

Normal blood Hb level

- Male : 14-18 gm/100ml
- Female : 12-16 gm/100ml
- In fetus : before birth: 16.5- 18.5 gm/100ml
- After birth: 23 gm/100ml

STRUCTURE OF HEMOGLOBIN

- Haemoglobin is a conjugated protein.
- It consist of a protein part and an iron containing part
- Protein part is globin

- Iron containing part is heme

(Heme present in muscle is called myoglobin)

(Heme present in brain is called neuroglobin)

Moecular weight: 68,000

Globin

- The protein globin, present in the Hb, is made of 4 polypeptide chain.
- According to sequence of amino acids in the primary structure of each chain, there are four types of chains; α, β, γ and δ.

Heme

- The heme is an iron- porphyrin complex.
- It consist of porphyrin nucleus and the iron

Porphyrin nucleus:

- It consist of 4 pyrrole rings.
- The pyrrole rings are joined by four methine bridges.

The iron:

The iron in the heme is ferrous (Fe2+) form.

The iron is attached to nitrogen atom of each pyrrole ring.

Attachment of heme to globin

- One molecule of Hb contains four units of haem, each attached to one of the four polypeptide chains constituting globin.
- As there are four units of heme in one molecule of Hb, so there are four iron atoms in one molecule of Hb which can carry four molecules (eight atoms) of oxygen.

FUNCTIONS OF HEMOGLOBIN

- Transport of oxygen
- Transport of carbon dioxide

Oxygen from the lungs to tissues.

- O2 +Hb → Oxyhemoglobin.
- The iron remains in ferrous state in this compound.
- Oxyhemoglobin is an unstable compound and the combination is reversible, i.e. when more oxygen is available, it combines with hemoglobin and whenever oxygen is required, hemoglobin can release oxygen readily .
- When oxygen is released from oxyhemoglobin, it is called reduced hemoglobin or ferrohemoglobin.

Transport of carbon dioxide

- CO2+Hb → carbaminohemoglobin
- It is also an unstable compound and the combination is reversible,
- i.e. the carbon dioxide can be released from this compound.
- The affinity of hemoglobin for carbon dioxide is 20 times more than that for oxygen

TYPES OF Hb (According to the difference in the globin chain)

1. Adult Hb (Hb A) = 2 α and 2 β subunits
   - HbA1 is the major form of Hb in adults and in children over 7 months.
   - HbA2 (2 α, 2 δ) is a minor form of Hb in adults. It forms only 2 – 3% of a total Hb A.
2. Fetal Hb (Hb F) = 2 α and 2 γ subunits

ABNORMAL HEMOGLOBIN

- The abnormal hemoglobin are produced because of structural changes in the polypeptide chains caused by mutation in the genes of the globin chains.
- It is categorized into:
  - Hemoglobinopathies
  - Hemoglobin in thalassemia and related disorders.

Hemoglobinopathies

Hemoglobinopathies are genetic disorders caused by abnormal polypeptide chains of hemoglobin.

Some of the hemoglobinopathies are:

Hemoglobin S - It is found in sickle cell anemia. In this, the α-chains are normal and β-chains are abnormal.

Hemoglobin C - The β-chains are abnormal. It is found in people with hemoglobin C disease, which is characterized by mild hemolytic anemia and splenomegaly.

Hemoglobin E - Here also the β-chains are abnormal. It is present in people with hemoglobin E disease which is also characterized by mild hemolytic anemia and splenomegaly.

Thalassemia - Thalassemia is an inherited disorder, characterized by abnormal hemoglobin.

Thalassemia is of two types:

- α-thalassemia
- β-thalassemia

α-Thalassemia

- In this α-chains are less, absent or abnormal.
- In adults, β-chains are in excess and in children, γ-chains are in excess.
- This leads to defective erythropoiesis and hemolysis.
- The infants may be stillborn or may die immediately after birth.

β-Thalassemia

In βthalassemia, βchains are less in number, absent or abnormal with an excess of α-chains.

The α-chains precipitate causing defective erythropoiesis and hemolysis.

Hemoglobin Derivatives

- **Oxyhemoglobin – Hb + O2**
- **Deoxyhemoglobin – Hemoglobin without Oxygen**
- **Carbamino hemoglobin – Hb + CO2**
- **Carbocyhemoglobin – Hb + CO**
- **Methemoglobin –** Iron molecule of hemoglobin is oxidized from normal ferrous state to ferric state.
- **Sulfhemoglobin -** Sulfhemoglobin is the abnormal hemoglobin derivative, formed by the combination of hemoglobin with hydrogen sulfide.
-

FATE OF RBC

- Senile erythrocytes (120 days old RBC) are destroyed in reticulo endothelial system and hemoglobin is released from them.
- Hemoglobin is broken into globin and heme.
- Globin enters into the amino acid pool.
- Heme is split into iron and porphyrin nucleus.
- Iron goes to iron pool and is reused (transferrin, ferritin) for Hb sysnthesis.
- Porphyrin nucleus degrades to form a pigment biliverdin is reduced to bilirubin.
- Bilirubin is released into blood from the reticuloendothelial cells.
- The bilirubin is transported to liver by the plasma protein, albumin.
- Bilirubin circulating in the blood is called free bilirubin or unconjugated bilirubin
- The free bilirubin is taken up by the liver cells
- In the liver, it is conjugated with glucuronic acid to form conjugated bilirubin
- Conjugated bilirubin is then excreted into intestine through bile

Fate of conjugated bilirubin

- In intestine, 80% of the conjugated bilirubin is converted into urobilinogen by intestinal bacteria.
- Remaining 20% of conjugated bilirubin from intestine is absorbed into blood and enters the liver through portal vein (enterohepatic circulation). From liver, it is reexcreted in bile
- About 5% of urobilinogen is excreted by kidney through urine. In urine, due to exposure to air, the urobilinogen is converted into urobilin by oxidation
- Some of the urobilinogen is excreted in feces as stercobilinogen. In feces, stercobilinogen is oxidized to stercobilin.

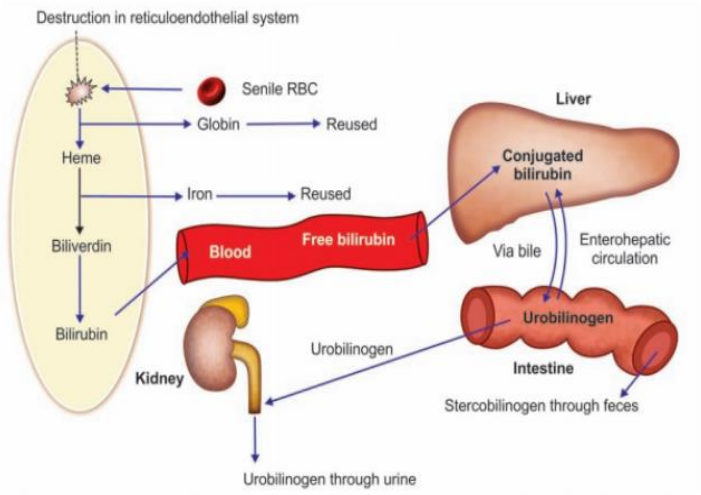
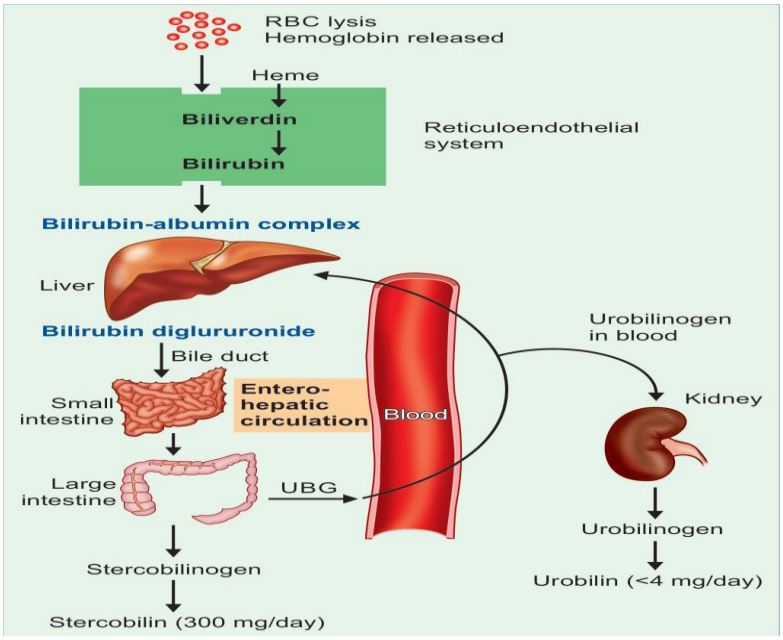


FIGURE 40.8: Formation and circulation of bile pigments



JAUNDICE

- Jaundice or icterus is the condition characterized by yellow coloration of the skin, mucous membrane and deeper tissues due to increased bilirubin level in blood.
- The normal serum bilirubin level is 0.2 to 0.8 mg/dL.
- Jaundice occurs when bilirubin level exceeds 2 mg/dL.

Types of Jaundice

- Pre-hepatic or hemolytic jaundice
- Hepatic or hepatocellular jaundice
- Post hepatic or obstructive jaundice

Pre-hepatic or Hemolytic Jaundice

- Hemolytic jaundice occurs because of excessive destruction of RBCs resulting in increased blood level of unconjugated bilirubin.
- In this condition, the excretory function of liver is normal.
- Unconjugated bilirubin increases enormously.
- The liver cells cannot excrete that much excess bilirubin rapidly.
- Unconjugated bilirubin is insoluble in water and is not excreted in urine.
- So, it accumulates in the blood resulting in jaundice.
- Formation of urobilinogen also increases resulting in the excretion of more amount of urobilinogen in urine.

Hepatic or Hepatocellular jaundice

- Hepatic jaundice occurs due to the damage of hepatic cells.
- Because of the damage, unconjugated and conjugated bilirubin is formed.
- The conjugated bilirubin from liver cannot be excreted and it returns to blood.

Post-hepatic or Obstructive

- Post hepatic jaundice occurs because of the obstruction of bile flow.
- The bile cannot be excreted into small intestine.
- So, bile salts and bile pigments enter the circulation.
- The blood contains more amount of conjugated bilirubin.

PCV (Packed Cell Volume)

- Packed cell volume (PCV) is the proportion of blood occupied by RBCs, expressed in percentage.
- It is the volume of RBCs packed at the bottom of a hematocrit tube when the blood is centrifuged.

Significance of determining PCV

- Diagnosis and treatment of anemia
- Diagnosis and treatment of polycythemia
- Determination of extent of dehydration and recovery from dehydration after treatment 4. Decision of blood transfusion.

Normal PCV:

- In males = 40% to 45%
- In females = 38% to 42%

PCV increases in:

- Polycythemia
- Dehydration

PCV decreases in:

- Anemia
- Cirrhosis of liver
- Pregnancy

BLOOD INDICES

Blood indices are the calculations derived from RBC count, hemoglobin content of blood and PCV.

Importance of blood indices

Blood indices help in diagnosis of the type of anemia.

Blood indices include:

- Mean corpuscular volume (MCV).
- Mean corpuscular hemoglobin (MCH).
- Mean corpuscular hemoglobin concentration (MCHC).
- Color Index (CI).
-

Mean Corpuscular Volume (MCV)

- MCV is the average volume of a single RBC and it is expressed in cubic microns (cu µ).
- Normal MCV is 90 cu µ (78 to 90 cu µ).
- When MCV is normal, the RBC is called normocyte.
- When MCV increases, the cell is known as a macrocyte and when it decreases, the cell is called microcyte.

$$MCV = \frac{\text{Hematocrit} \times 10}{\text{RBC count in million/cu mm of blood}}$$

$$= \frac{45 \times 10}{5}$$

$$= 90 \text{ Cu } \mu$$

Mean Corpuscular Hemoglobin (MCH)

- MCH is the quantity or amount of hemoglobin present in one RBC.
- It is expressed in micromicrogram or picogram (pg).
- Normal value of MCH is 30 pg (27 to 32 pg).

$$MCH = \frac{\text{Hb g/dl}}{\text{PCV/100}} \times 10$$

$$= \frac{15}{5} \times 10$$

$$= 30 \text{ pg}$$

Mean Corpuscular Hemoglobin Concentration (MCHC)

- MCHC is the concentration of hemoglobin in one RBC.
- It is the amount of hemoglobin expressed in relation to the volume of one RBC.
- So, the unit of expression is percentage.
- This is the most important absolute value in the diagnosis of anemia. Normal value of MCHC is 30% (30% to 38%).
- When MCHC is normal, the RBC is normochromic. When the MCHC decreases, the RBC is known hypochromic.

$$\text{MCHC} = \frac{\text{Hb g/dl}}{\text{PCV in 100 ml}} \times 100$$

$$= \frac{15}{45} \times 100$$

$$= 33.3\%$$

Color Index (CI)

- Color index is the ratio between the percentage of hemoglobin and the percentage of RBCs in the blood.
- Actually, it is the average hemoglobin content in one cell of a patient compared to the average hemoglobin content in one cell of a normal person.
- Normal color index is 1.0 (0.8 to 1.2).
- It increases in macrocytic (pernicious) anemia and megaloblastic anemia. It is reduced in iron deficiency anemia. And, it is normal in normocytic normochromic anemia.

$$\text{CI} = \frac{\text{Hb \%}}{\text{RBC \%}}$$

ANEMIA

Anemia is defined as decreased red cell count or hemoglobin content of blood. It is a disorder, characterized by the reduction in:

- Red blood cell (RBC) count
- Hemoglobin content
- Packed cell volume (PVC).

Classification of Anemia

- Morphological Classification
- Etiological Classification

Morphological Classification:

- Normocytic Normochromic anemia - Size (MCV) and color (MCHC) of RBCs are normal. But the number of RBC is less
- Macrocytic Normochromic anemia- RBCs are larger in size with normal color. RBC count is less.
- Macrocytic Hypochromic anemia - RBCs are larger in size. MCHC is less, so the cells are pale (less colored).
- Microcytic Hypochromic anemia - RBCs are smaller in size with less color.

Etiological Classification:

I. Hemorrhagic anemia
II. Hemolytic anemia
III. Deficiency anemia
IV. Aplastic anemia
V. Anemia of chronic diseases.

I. Hemorrhagic anemia

- Occurs due to excess blood loss.

Two types:

Acute hemorrhagic anemia - Acute hemorrhage refers to sudden loss of a large quantity of blood as in the case of accident.

Chronic hemorrhagic anemia - It refers to loss of blood by internal or external bleeding, over a long period of time. It occurs in conditions like peptic ulcer, purpura, hemophilia and menorrhagia.

II. Hemolytic Anemia

- Hemolysis means destruction of RBCs.
- Anemia due to excessive hemolysis which is not compensated by increased RBC production is called hemolytic anemia.

Hemolytic anemia's are classified into:

a. Hereditary hemolytic anemia
b. Acquired hemolytic anemia

Hereditary Hemolytic anemia's are:

Sickle cell anemia –

- This is a hereditary disorder in which red cells contain an abnormal hemoglobin called Hb S.
- Hb S is the hemoglobin in which glutamic acid is replaced by valine at the 6th position of beta chain. In the deoxygenated state, conformational changes induced by Hb S makes the cell more rigid and deformed to take the shape of a sickle.
- Therefore, cells undergo intravascular hemolysis.

Thalassemia –

- Thalassemia is a genetically determined heterogenous group of diseases in which the rate of synthesis of one or more types of hemoglobin polypeptide chain is decreased. This results in the hemolysis.

Hereditary spherocytosis –

Spherocytes have a decreased surface area to volume ratio and the cells are more rigid (less deformable). Therefore, when cells pass through the splenic pulp, they are destroyed.

Acquired haemolytic anaemias are:

- *Immunohaemolytic anaemia (due to antibodies against RBCs),*
- *Haemolytic anaemia due to direct toxic effects (e.g. in malaria, snake venom, toxic effects of drugs and chemicals),*
- *Haemolytic anaemia in splenomegaly*

III. Aplastic anaemia

- Aplastic anemia is due to the disorder of red bone marrow.
- Red bone marrow is reduced and replaced by fatty tissues.
- Bone marrow disorder occurs in the following conditions:

  a. Repeated exposure to Xray or gamma ray radiation.
  b. Presence of bacterial toxins, quinine, gold salts, benzene, radium, etc.
  c. Tuberculosis.
  d. Viral infections like hepatitis and HIV infections.

IV. Anemia due to chronic diseases

- Rheumatoid Arthritis
- Tuberculosis

V. Deficiency Anemia

- Iron deficiency anemia - Anemia of IDA is microcytic hypochromic type.
- Megaloblastic anemia due to deficiency of Vit B12-
- Megaloblastic anemia due to deficiency of folic acid deficiency
- Anemia due to deficiency of proteins and Vitamin C.

WHITE BLOOD CELLS

White blood cells (WBCs) or leukocytes are the colorless and nucleated formed elements of blood (leuko is derived from Greek word leukos = white).

Normal count:

The normal count of WBC in an adult is 4000 – 11,000/mm3.

Variations in WBCs count

<u>Leucocytosis</u>

Leucocytosis refers to increase in total WBC count above 11,000/mm3 .

<u>Leucopenia</u>

Leucopenia refers to decrease in the total WBC count below 4000/mm3 .

Leukaemia

Leukaemia is a malignant disease of the blood in which there occurs an uncontrolled proliferation of total WBC count associated with the presence of immature WBCs in the peripheral blood.

Total WBC count is usually above 50,000/mm3 , and may be as high as 100,000–300,000/ mm

WBC ‘s are classified into:

- Granulocytes
- Agranulocytes

Granulocytes

Granulocytes have granules in its cytoplasm.

- Neutrophil
- Eosinophil
- Basophil

Agranulocytes

Devoid of granules in the cytoplasm.

- Monocytes
- Lymphocytes

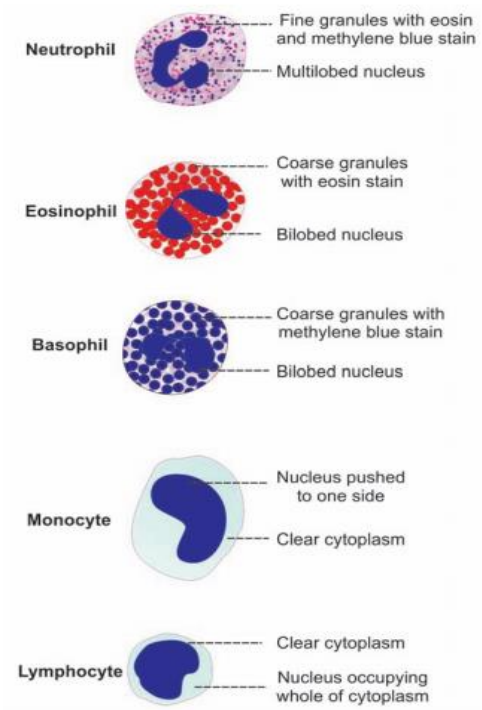


FIGURE 16.1: Different white blood cells

Ms. SNEHA BABY
LECTURER
DEPARTMENT OF PHYSIOLOGY

Neutrophil

General features:

- Size: 10 – 14 µm
- Site of production – Bone marrow
- Life span – 2 – 7 Days

Nucleus

- Horse-shoe shaped nucleus.
- Nucleus is purple in colour.
- They are multilobed.
- The lobes of the nucleus are connected by chromatin filaments.

Cytoplasm

- Pale bluish in colour
- Fine granules are present.
- Granules take both acidic and basic stains and look violet-pink in colour. (When stained with leishman's stain)
- Granules are lysosomal in origin
- The granules also liberate histamines and peroxidase enzyme.

Functions:

- Phagocytosis

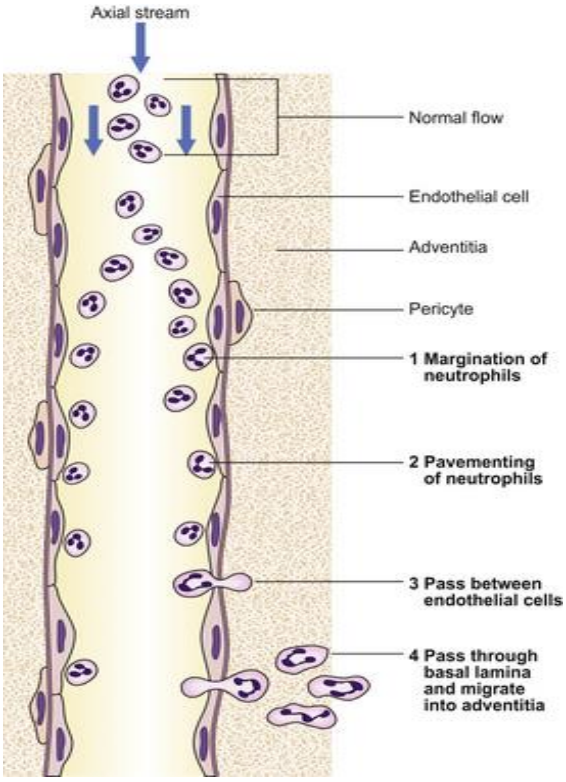
Phagocytocis

It involves:

- Margination
- Emigration and diapedisis
- Chemotaxis
- Opsonization
- Engulfment
- Secretion
- Killing or degradation

Margination

- In the area of infection the neutrophills get marginated.
- It get attached to the capillary endothelium and start rolling along its surface.



Emigration and diapedesis

- The marginated neutrophils are emigrated from the blood to the site of *infection by diapedesis into the tissues by passing through* the junction between endothelial cells of the blood vessels.

Chemotaxis

- It is the process by which the neutrophils are attracted towards bacteria at the site of inflammation.

Opsonization

- Opsonization refers to the process of coating of bacteria by the *opsonins by which* bacteria become tasty to the phagocytes.

Engulfment stage

- The neutrophils project *pseudopodia* in all directions around the opsonized particle which is bound to the surface of neutrophil.
- Pseudopodia meet each other on opposite side and fuse.
- This creates an enclosed chamber with the engulfed material.
- It breaks away from the membrane forming a *phagocytic vesicle. Then the lysosomes of the cell fuse with the* phagocytic vesicle to form *phagolysosome or phagosome.*

Secretion and degranulation

- Once the bacterium is engulfed, the lysosomes pour their enzymes into the vesicle and also in interstitial space .This process is called degranulation.
- There are large numbers of proteolytic enzymes especially geared up for digesting bacteria.
- In addition, lysosomes of macrophages also contain lipases which can digest the thick lipid membranes possessed by certain bacteria.

Killing or degradation stage

- The neutrophils and macrophages contain *bactericidal agents (defensins α and β)* which can kill most of the bacteria even if the lysosomal enzyme fails to digest them.

<u>Eosinophil</u>

General Features:

- Size- 10–14 μm.
- Site of production – Bone marrow
- Life span – 7 – 12 days

Nucleus:

- Nucleus is purple in colour and is bilobed
- The two lobes are connected by the chromatin strands and thus look *spectacle shaped.*

Cytoplasm:

- Cytoplasm is acidophilic
- Appears bright pink in colour.
- It contains coarse, deep red staining granules
- The granules contain histamine, lysosomal enzymes.

Functions:

1. ***Mild phagocytosis***- Eosinophils are not very motile so very mild phagocytic activity.
2. ***Role in parasitic infestations*** - They play an important role in the defence mechanism of body especially *in parasitic infestations.*
3. ***Role in allergic reaction*** - The eosinophils increase in allergic conditions like bronchial asthma and hay fever.
4. They are capable of *detoxifying* inflammation inducing substances like histamine and bradykinin.

<u>Basophils</u>

- Size - 10–14 µm.
- Size of production – Bone Marrow
- Life span - 12 – 15 days

Nucleus:

- Nucleus of basophils is irregular
- Bi-lobed or tri-lobed
- Boundary is not clearly defined because of overcrowding with the
- coarse granules.

Cytoplasm:

- Cytoplasm of basophils is slightly basophilic and appears blue.
- It is full of Coarse granules

Functions:

1. *Mild phagocytosis*. Basophils have very mild phagocytic function.

2. *Role in allergic reaction*. Basophils release histamine,bradykinin,

slow reacting substances of anaphylaxis and serotonin (5HT).

These substances, in turn, cause local vascular and tissue reactions that

cause many allergic manifestations.

3. *Role in preventing spread of allergic inflammatory process.*

Monocytes

- Size - 12–20 µm
- Site of Production – Bone Marrow
- Life span – 2 – 5 days

Nucleus:

- The nucleus of a monocyte is large, single and eccentric in position, i.e. present on one side of the cell.
- It is kidney shaped.

Cytoplasm:

- The cytoplasm is abundant, pale blue and usually clear (no granules)

Functions

- **Role in defence mechanism**. Monocytes along with the neutrophils play an important role in the body's defence mechanism. Their main function *is phagocytosis.*
- **Role in tumour immunity**. Monocytes may also kill tumour cells after sensitization by the lymphocytes.
- **Synthesis of biological substances**. Monocytes synthesize complement and other biologically important substances.

Lymphocytes

There are two types of lymphocytes, large and small

Small Lymphocyte

- Size – 7 - 10 µm
- Site of production – Bone Marrow
- Life span – half to 1 day

Large lymphocytes

- Size – 12- 16 µm
- Site of production – Bone Marrow
- Life span -

Nucleus:

- Lymphocytes have a large round, single nucleus which almost completely fills the cell.
- It stains blue very deeply giving *ink-spot appearance.*

Cytoplasm:

- The cytoplasm is scanty, i.e. its amount is always less than that of the nucleus.

- It is seen as a crescent of clear light blue colour around the nucleus.
- Cytoplasm does not contain visible granules

Functional subtypes

- Based on their developmental background, life span and functions, the small lymphocytes have been broadly classified into three subtypes .

B lymphocytes- bone marrow and concerned with the humoral immunity.

T lymphocytes - thymus and concerned with cellular immunity.

Natural killer (NK) cells are lymphocyte-like cells that non-specifically kill any cell that is coated with IgG.

Thus NK cells provide *innate immunity.*

Functions

- B lymphocytes - Responsible for the development of *humoral immunity* also called antibody-mediated immunity .
- T lymphocytes - Responsible for the development of *cellular immunity*, also called cell-mediated immunity or T cell immunity.

FUNCTIONS OF LEUKOCYTES

- Neutrophils forms the first line of defense and the major function is the Phagocytosis.
- Eosinophils have role in mild phagocytosis, allergic reactions also defend against the parasitic infestations with the help of Major basic proteins, eosinophil cationic proteins etc.
- Basophils provide immunity against the allergic responses by releasing histamines, heparin, hyaluronic acid etc…which are present in the granules.
- Monocytes along with neutrophils forms the first line of defense and shows phagocytic responses.
- Lymphocytes provide cellular immunity and humoral immunity.

****

IMPORTANT QUESTIONS

10 Marks Questions

1. Erythropoiesis and factors affecting.
2. Explain anemia and its classification.

5 Marks Questions

1. Composition and functions of blood.
2. Functions of Hemoglobin.
3. Fate of RBC
3. Jaundice
4. Explain anemia and its classification.
5. Blood indices
6. Phagocytosis

3 Marks Questions

1. Functions of blood.
2. Functions of RBC.
3. Function of Hemoglobin.
4. Derivatives of Hemoglobin.
5. Hemoglobinopathies.
6. Functions of leukocytes.
7. Calculate MCV/MCH/MCHC/CI.
8. Blood indices.
9. Jaundice.
10. Phagocytosis.